

Quantum — Entity Relationship Diagram

Data model for the Quantum project management application · MongoDB / Mongoose

PK

Primary Key — unique document identifier (`_id`)

FK

Foreign Key — references another entity's `_id`

UK

USER	PROJECT	TASK
<code>_id</code> ObjectId PK	<code>_id</code> ObjectId PK	<code>_id</code> ObjectId PK
<code>name</code> String	<code>name</code> String	<code>title</code> String
<code>username</code> String UK	<code>description</code> String	<code>description</code> String
<code>email</code> String UK	<code>status</code> active archived	<code>status</code> To Do In Progress Complete
<code>password</code> String (hashed)	<code>owner</code> ObjectId FK	<code>project</code> ObjectId FK
<code>createdAt</code> Date	<code>members</code> ObjectId[] FK	<code>owner</code> ObjectId FK
<code>updatedAt</code> Date	<code>createdAt</code> Date	<code>createdAt</code> Date
	<code>updatedAt</code> Date	<code>updatedAt</code> Date

RELATIONSHIPS

USER

1

→

many

PROJECT

— A user owns many projects. `Project.owner` references `User._id`

PROJECT

1

→

many

TASK

— A project contains many tasks. `Task.project` references `Project._id`

USER

1

→

many

TASK

— A user creates many tasks. `Task.owner` references `User._id`

PROJECT

many

↔

many

USER

— A project can have many collaborators. `Project.members` is an array of `User._id` refs